

# Python Programming Course Outline

Duration: 6-10 Weeks

Prerequisites: None (designed for complete beginners)

## Module 1: Introduction to Python

- What is Python and why use it?
- Installing Python & IDEs (IDLE, VS Code, Jupyter)
- Running your first Python script
- Python syntax and code structure
- Understanding the Python interpreter

## Module 2: Python Basics

- Variables and Data Types (int, float, str, bool)
- Basic input/output functions
- Arithmetic and comparison operators
- Comments and indentation
- Type casting

## Module 3: Control Flow

- Conditional Statements (if, elif, else)
- Loops: for and while
- Loop control: break, continue, pass
- Nested loops

## Module 4: Functions and Modules

- Defining and calling functions
- Parameters and return values
- Built-in functions vs user-defined functions
- Variable scope
- Importing modules and using standard libraries (math, random, etc.)

## Module 5: Data Structures

- Lists: creation, indexing, slicing, methods
- Tuples: immutability and use cases
- Dictionaries: key-value pairs, operations
- Sets: uniqueness and operations
- List/dictionary comprehensions

## **Module 6: Working with Strings**

- String creation and formatting
- Common string methods
- f-strings and format()
- String indexing and slicing

## **Module 7: File Handling**

- Opening and reading files
- Writing and appending to files
- Working with text and CSV files
- Using with context manager

## **Module 8: Error Handling and Exceptions**

- Common Python errors and exceptions
- Try, except blocks
- Finally and else in exception handling
- Raising exceptions

## **Module 9: Object-Oriented Programming (OOP)**

- Introduction to classes and objects
- Attributes and methods
- Constructors (\_\_init\_\_)
- Inheritance and method overriding
- Encapsulation and private attributes

## **Module 10: Python Libraries Overview**

- Introduction to popular Python libraries:
- - datetime

- - os, sys
- - collections
- - json
- Brief intro to pandas, numpy, and matplotlib for future data science use

## **Module 11: Introduction to Virtual Environments & PIP**

- Installing and using pip
- Creating virtual environments
- Installing third-party libraries

## **Module 12: Final Project**

- Plan and build a small project (CLI or script)
- - Examples: calculator, to-do app, weather app, quiz game, file analyzer
- Apply all learned concepts
- Present and document your code

## **Optional / Advanced Topics**

- Regular Expressions with re
- Lambda, map, filter, reduce
- Recursion
- Introduction to testing with unittest
- Introduction to Web Development with Flask
- GUI Programming with Tkinter